Name ____ Date ___________ Period _____

Continuous vs. Discrete

Algebra II | Unit 1 | Day 3 | TEKS A2.2.A, A2.7.I

Objective: I can decide whether a domain and range are continuous or discrete, and write each one using the notation that fits.

Vocabulary

Continuous function — the graph is _________________ — you could trace it without lifting your pencil.

Discrete function — the graph is made of separate, ___________________ points.

Part 1 — Sort the Situations

Write C for continuous or D for discrete.

Situation	C or D	Why?
The number of chairs in a classroom	_____	____________________
The distance a car travels	_____	____________________
Your age measured in years and days	_____	____________________
The number of pets a family owns	_____	____________________
The weight of a watermelon	_____	____________________

Part 2 — The Same Function, Two Ways

A cube-shaped box has side x inches, so its volume is **$V(x) = x^3$**.

x	1	2	3	4
$V(x) = x^3$	1	8	27	64

Case A: the box can be any size at all.
Domain __________________________ Range __________________________ _______________

Case B: the box is built out of 1-inch cube blocks, so x must be a whole number.
Domain __________________________ Range __________________________ _______________

Same equation, different _______________ → different domain. The context decides.

Part 3 — Which Notation Fits?

Continuous domains use _______________ notation, like [0, 5].

Discrete domains get _______________ in braces, like { 0, 1, 2, 3 }, because there are gaps between the values.

You Try 1. A stadium sells 0 to 5 hot dogs per person. Domain __________________________

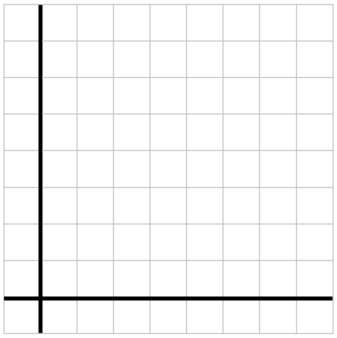
You Try 2. A candle burns from 10 cm down to 0 cm. Domain of heights ____________________

You Try 3. Why can You Try 1 not use interval notation? ________________________________

Part 4 — Graph Both Kinds

Each square is 1 unit. Plot each, then name the domain and range.

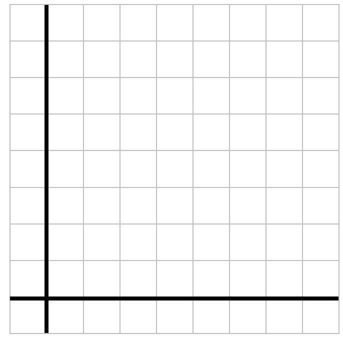
A. Discrete. A tutor charges $2 per session, up to 4 sessions. Plot (1, 2), (2, 4), (3, 6), (4, 8).



Domain ___________________

Range ___________________

B. Continuous. A candle burns 2 cm per hour from 8 cm. Graph $h = 8 - 2t$ for $0 \le t \le 4$.



Domain ___________________

Range ___________________

Part 5 — Quick Check

1. Should a discrete domain ever be written as an interval? ____________________________
2. Can a continuous range include negative numbers? ____________________________
3. Is money spent at a vending machine continuous? ____________________________

Practice

Decide continuous or discrete, then write the domain in the notation that fits.

1. A runner covers any distance from 0 to 5 miles.

C or D ________ Domain __________________________ Why? ____________________________

2. A vending machine sells 0 to 6 drinks.

C or D ________ Domain __________________________ Why? ____________________________

3. Water in a bathtub drops from 30 gallons to 0.

C or D ________ Range __________________________ Why? ____________________________

4. A class of 24 students is split into equal teams of 4.

C or D ________ Domain (teams) __________________________

5. Explain in your own words why a discrete domain cannot use interval notation.

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Exit Ticket

A school orders pizzas for a party. Each pizza feeds 6 students, and they order up to 5 pizzas.

Domain (pizzas) __________________________ Range (students fed) ____________________________

Continuous or discrete, and why? _____________________________________